MILITARY SPECIFICATION SHEET

PLASTIC SHEETS, LAMINATED, THERMOSETTING ELECTRICAL-INSULATING

SHEET, POLYESTER GLASS-MAT GRADE GPO-N1 (CLASSES 130, 155, and 180)

The complete requirements for procuring the plastic sheets described herein shall consist of this document and the latest issue of Specification MIL-P-24364(SHIPS)

The plastic laminated sheets shall be as follows:

Grade GPO-N1, Polyester Resin Glass-Mat, for Mechanical Applications

Requirements: The property values shall be as shown in tables I, II, III, and IV.

FSC 9330

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Table I - Property value requirements for

Qualification Tests for Polyester Resin Glass-Mat Grade GPO-N1

CLASS 130

Properties to be tested	Test Paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Units of value	Values required for each thickness 1/16 inch	1/8 inch	1/2 inch
Thickness	4.7.3	1	as received	inch	± 0.008	± 0.010	± 0.024
Dielectric breakdown parallel to laminations	4.7.5	1 (short-time)					
		4 (step-by-step)	as received	average KV-min.	25	40	40
		1 (short-time)					
		4 (step-by-step)	336/50 water	average KV-min.	15	25	30
Dielectric constant	4.7.6			maximum			
60 Hertz		4	as received		---	5.0	---
		4	24/23 water		---	6.0	---
1 megahertz		4	as received		---	4.5	---
		4	24/23 water		---	5.5	---
Dissipation factor	4.7.6			maximum			
60 hertz		4	as received		---	0.05	---
		4	24/23 water		---	0.15	---
1 megahertz		4	as received		---	0.03	---
		4	24/23 water		---	0.06	---
ARC resistance	4.7.7	4	as received	seconds, min.	100	100	100
		4	48/50 water		100	100	100
Wet tracking	4.7.8	4	as received	minutes, min.	95 1/	95 1/	95
Impact strength, tested edgewise	4.7.9	4	48/50	foot lbs. per inch-min.		8	8
Flexural strength, tested flatwise (crosswise)	4.7.10	5	as received	P.S.T.-min.	18000	18000	18000
Bond strength	4.7.11	4	as received	lbs.-min.	---	---	850
		4	48/50 water		---	---	800
Water absorption	4.7.12	4	1/105+ 24/23 water	percent-max.	0.8	0.5	0.3
Thermal endurance	4.7.14			Temp. index °C at 25,000 hours.-min.	---	130	---
Silicone content	4.7.15	1	168/185	ppm-max.	25	25	25
Machinability	4.7.16	1	as received	pass or fail	---	----	pass

1/ Specimens of the same material shall be built up to 0.250 inch.

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Table II - Property value requirements for

Qualification Tests For Polyester Resin Glass-mat Grade GPO-N1

CLASS 155.

Properties to be tested	Test paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Unit of value	Values required for each thickness		
					1/16 inch	1/8 inch	1/2 inch
Thickness	4.7.3	1	as received	inch	± 0.008	± 0.010	± 0.024
Dielectric breakdown parallel to laminations	4.7.5	1 (short-time)					
		4 (step-by-step)	as received	average KV-min.	25	40	40
		1 (short-time)					
		4 (step-by-step)	336/50 water	average KV-min.	15	25	30
Dielectric constant	4.7.6			maximum			
60 hertz		4	as received		---	5.0	---
		4	24/23 Water		---	6.0	---
1 megahertz		4	as received		---	4.5	---
		4	24/23 Water		---	5.5	---
Dissipation factor	4.7.6			maximum			
60 hertz		4	as received		---	0.05	---
		4	24/23 Water		---	0.15	---
1 megahertz		4	as received		---	0.03	---
		4	24/23 Water		---	0.06	---
ARC resistance	4.7.7	4	as received		100	100	100
			48/50 Water	seconds-min.	100	100	100
Wet tracking	4.7.8	4	as received	minutes-min.	95 1/	95 1/	95
Impact strength, tested edgewise	4.7.9	4	48/50	foot lbs. per inch-min.	---	8	8
Flexural strength, tested flatwise, (crosswise)	4.7.10	5	as received	P.S.I.-min.	18000	18000	18000
Bond strength	4.7.11	4	as received	lbs.-min.	---	---	850
		4	48/50 Water		---	---	800
Water absorption	4.7.12	4	1/105+ 24/23 Water	percent-max.	0.8	0.5	0.3
Thermal endurance	4.7.14			Temp. index°C at 25,000 hrs.-min.	---	155	---
Silicone content	4.7.15	1	168/185	P.P.M-max.	25	25	25
Machinability	4.7.16	1	as received	pass or fail	---	---	pass

1/ Specimens of the same material shall be built up to 0.250 inches.

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Table III - Property Value Requirements For

Qualification Tests For Polyester Resin Glass-Mat GPO-N1,

CLASS 180.

Properties to be tested	Test paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Units of value	values required for each thickness 1/16 inch	1/8 inch	1/2 inch
Thickness	4.7.3	1	as recieved	inch	± 0.008	± 0.010	± 0.024
Dielectric breakdown parallel to laminations	4.7.5	1 (short-time)					
		4 (step-by-step)	as received	average KV-min.	25	40	40
		1 (short-time)					
		4 (step-by-step)	336/50 Water	average KV-min.	15	25	30
Dielectric constant	4.7.6			maximum			
60 hertz		4	as received		---	5.0	---
		4	24/23 Water		---	6.0	---
1 megahertz		4	as received		---	4.5	---
		4	24/23 Water		---	5.5	---
Dissipation factor	4.7.6			maximum			
60 hertz		4	as received		---	0.05	---
		4	24/23 Water		---	0.15	---
1 megahertz		4	as received		---	0.03	---
		4	24/23 Water		---	0.06	---
ARC resistance	4.7.7	4	as received	seconds-min.	100	100	100
			48/50 Water		100	100	100
Wet tracking	4.7.8	4	as received	minutes-min.	95 1/	95 1/	95
Impact strength, tested edgewise	4.7.9	4	48/50	foot lbs per inch-min.		8	8
Flexural strength, tested flatwise, (crosswise)	4.7.10	5	as received	P.S.I.-min.	18000	18000	18000
Bond strength	4.7.11	4	as received	lbs.-min.	---	---	850
		4	48/50 Water		---	---	800
Water absorption	4.7.12	4	1/105+ 24/23 Water	percent-max.	0.8	0.5	0.3
Thermal endurance	4.7.14			Temp. Index °C. at 25,000 hrs.-min.	---	180	---
Machinability	4.7.16	1	as received	pass or fail	---	---	pass

1/ Specimens of the same material shall be built up to 0.250 inch.

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Table IV - Property value requirements for

quality conformance tests for polyester resin glass-mat grade GPO-N1 - ALL CLASSES.

Properties to be tested	Test paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Unit of value	Values required for each thickness range[1/] - (Inches)				
					0.0625 or less	0.0626-0.094	0.095-0.250	0.251-1.000	1.001-2.000
Warp or twist	4.7.4	1	as received	percent-max.	2.50%	---	2.50%	0.50%	---
Dielectric breakdown parallel to laminations	4.7.10	1 (short-time) 4 (step-by-step)	48/50/Water	average KV-min.	15	15	25	30	35
Flexural strength, tested flatwise, (crosswise)	4.7.1	4	as received	P.S.I.-min.	16000	16000	18000	16000	14000
Bond strength	4.7.11	4	as received	lbs.-min.	---	---	---	850[2/]	850

[1/] These ranges applicable for the nominal thickness.
[2/] For specimens 0.500 inch and above.

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